Hedging Against Lag Order Uncertainty in SVARs: Bayesian Model Averaging versus Information Criteria

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Abstract

Structural Vector Autoregressions (VARs) identified via set restrictions are highly sensitive to lag order specification. Updating Ivanov and Kilian (2005), we evaluate traditional Information Criteria against Bayesian Model Averaging (BMA) in recovering the true median target impulse response function (IRF). Using Monte Carlo simulations calibrated to global oil market data, we compare information criteria against several sppecifications of Bayesian Model Averaging (BMA) in recovering structural shocks. For single-model selection, the Akaike Information Criterion (AIC) consistently dominates stricter criteria by avoiding severe underfitting bias. However, a joint-grid BMA approach—averaging across lag orders and BVAR shrinkage hyperparameters—yields the most accurate IRF recovery, outperforming simple AIC in almost all scenarios. By dynamically shrinking and weighting a wide set of lag order specifications, this method mitigates lag uncertainty while preserving lower-order precision, offering a robust alternative to traditional single-model selection.

Keywords: Structural Vector Autoregression (SVAR), Set Identification, Bayesian Model Averaging (BMA), Model Uncertainty, Lag Order Selection, Monte Carlo Simulation

JEL Codes: C11, C15, C32, C52

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This paper has been completed in conjunction with Aydın (2026). The main difference is that this paper extends the method space by introducing BMA, whereas Aydın (2026) implements BVAR. Therefore, both the last subsection of the theoretical introduction, Subsection D, as well as Section 4 and Section 5 differ. Replication files are available in the GitHub repository for both papers.

1 Introduction

Structural Vector Autoregressions (VARs) remain the cornerstone of a large part of empirical macroeconomics, especially for monetary policy, shock transmission, and financial markets. While forecasting may be the main goal for some researchers, many economists also use it to estimate the relationships among factors that determine macroeconomic movements. For this purpose, Impulse Response Functions (IRFs) are derived from the VAR parameters, establishing how a shock in one variable affects the development of another period, both contemporaneously and over horizons of several months or even years. Using this transformation, structural VARs are continually adapted to answer complex questions, such as how speculative trading and real-time news shocks drive global oil market dynamics (Kilian and Murphy, 2014; Gazzani et al., 2024; Castelnuovo et al., 2024), how high-frequency monetary policy surprises transmit through corporate credit spreads (Gertler and Karadi, 2015; Ferreira et al., 2023), and how these policy shifts impact global commodity prices (Miranda-Pinto, 2023).

The specification of the VAR model, as well as the method to decompose it into its structural form, are the two crucial specifications that a researcher has to decide on. Traditionally, VAR models were specified with a lag order determined by economic theory or an information criterion (IC) and then estimated by Ordinary Least Squares (OLS) or Maximum Likelihood (ML). The past 15 years have seen a surge in the popularity of Bayesian methods, in which not a single lag order is specified, but lag orders are weighted and/or shrunk based on a prior and a posterior probability distribution (Blake and Mumtaz, 2015). These methods enable the researcher not only to rely on a single specification but also to incorporate a larger model or set of models while still keeping variance low. The methodology for estimating IRFs has also advanced significantly since Sims (1980) showed that structural IRFs can be retrieved via Cholesky decomposition, i.e., a predetermined causal ordering of the relationships between variables. Newer methods typically require weaker assumptions and adopt a more agnostic approach to inferring economic relationships. The dominant approach nowadays is set identification using sign restrictions informed by economic theory (Uhlig, 2005; Rubio-Ramírez et al., 2010). Unlike in Cholesky decomposition, all variables can simultaneously react to each other's shocks, and sign restrictions can reduce the set of possible IRFs to those that fit the researcher's assumptions. This yields a set of admissible IRFs. Typically, the median target—the specific IRF closest to the point-wise median of the admissible set—is selected as the primary result (Fry and Pagan, 2011).

This study draws inspiration from Ivanov and Kilian (2005), the first paper to examine which VAR specification best retrieves the true IRF using Monte Carlo simulations. In 2005, this meant testing criteria for lag selection to assess how well they fit VAR models and retrieve structural IRFs via Cholesky decomposition. We aim to update this analysis to the more recent world of VAR methodology, where IRFs are identified via sign restrictions and estimation relies heavily on Bayesian methods rather than solely on information criteria.. We construct a realistic DGP based on Kilian and Murphy (2014)'s study of the oil market, using the set identification method. Then we compare different methods across repeated simulations by how well they can retrieve the true IRF, using the median target of their identified set.

We then compare the information criteria AIC, BIC, and HQC with estimation using Bayesian Model Averaging (BMA) which doesn't fix one lag order but weights a set of probable lag orders and can be combined with Bayesian Vector Autoregression (BVAR) to shrink down parameters of higher lag orders. Our results demonstrate that hedging lag selection risk through Bayesian Model Averaging (BMA) significantly improves the accurate recovery of median target Impulse Response Functions (IRFs). Particularily the joint-grid BMA approach dominates traditional AIC-selected models in most scenarios and is shown to diversify estimation weights across the most broad spectrum of lag orders. By dynamically applying tighter hyperparameter shrinkage (λ_1) to overparameterized specifications ($p > p_0$), this approach successfully mitigates the risk of lag order uncertainty while preserving precise parameter estimation for the crucial lower lags. Furthermore, our results confirm the results presented by Ivanov and Kilian (2005) when relying on single-model selection. In this case, the Akaike Information Criterion (AIC) consistently outperforms stricter criteria like BIC and HQC, as its milder penalization avoids the severe underfitting bias that heavily distorts IRF inference. As T grows, AIC also improves in comparison to BMA for a sufficiently low p_0 as this enables parameters to be estimated with less variance also in simple OLS.

The rest of the paper is structured as follows: Section 2 introduces the reader to the theoretical foundations needed to understand the methodology of this simulation study. It first introduces the VAR framework, Section 2.1, then explains the sign restriction approach for IRF decomposition, Section 2.2, and explains information criteria as well as the BMA method, Section 2.3 and 2.4. We aim to both convey the intuition behind these methods as well as derive key properties. Section 3 explains the simulation design and Section 4 presents our results. Finally, the conclusion, Section 5, summarizes and discusses our key findings.

2 Theoretical Background

2.1 VAR Framework and IRFs

The reduced form of a vector autoregression, VAR(p_0), establishes a relationship between current observations of a set of variables with the past observations of these variables up until a pre-determined lag p_0:

$$Y_t = v + A_1 Y_{t-1} + A_2 Y_{t-2} + ... + A_{p_0} Y_{t-p_0} + e_t \tag{1}$$

Here, Y_t is a $(K \times 1)$ vector of endogenous variables, v is a vector of intercepts, $A_1, \ldots, A_{p_0}$ are $(K \times K)$ coefficient matrices, e_t is a white-noise innovation vector with $\mathbb{E}[e_t] = 0$ and covariance matrix $\mathbb{E}[e_t e_t'] = \Sigma$, and p_0 denotes the lag order. This is the standard reduced-form representation of a VAR process (Lütkepohl, 2005; Sims, 1980). However, reduced-form innovations are composed of structural shocks and are therefore uninformative about the effects of any individual shock. To identify the structural shocks of interest, restrictive assumptions must be imposed. The relationship between the two is given by:

$$e_t = B\epsilon_t \tag{2}$$

where ϵ_t denotes the vector of mutually uncorrelated structural shocks (that are normalized to unit variance, i.e., $\mathbb{E}[\epsilon_t \epsilon_t'] = I$) and B is a matrix satisfying $\Sigma = BB'$. Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D, the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B. The literature offers several strategies for this purpose, including short-run (recursive) restrictions (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. Following Ivanov and Kilian (2005), we do not commit to a single scheme mentioned above but instead adopt, for each data-generating process, the identifying assumptions used in the original study on which it is based.

In general, interpreting the dynamics implied by VAR coefficients is difficult. Therefore, practitioners use Impulse Response Functions (IRF) to summarize the main dynamics that they want to explain. To obtain the coefficient set that constructs the impulse response function, we express the VAR model in terms of an infinite VMA process. Using (1):

$$Y_t = [I - A_1 L - A_2 L^2 - ... - A_{p_0} L^{p_0}]^{-1} v + [I - A_1 L - A_2 L^2 - ... - A_{p_0} L^{p_0}]^{-1} e_t \tag{3}$$

Which is equivalent to:

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i \, e_{t-i} \quad \text{and} \qquad \Phi_0 = I_K \tag{4}$$

where $\mu = [I - A_1 - \cdots - A_{p_0}]^{-1} v$ is the unconditional mean vector and the moving average coefficient matrices are obtained recursively as $\Phi_i = \sum_{j=1}^{i} A_j \Phi_{i-j}$ where $A_j =$

0 for $j > p_0$ (see appendix for detailed derivation). Finally, in order to obtain the structural impulse response functions, we plug the expression given in (2) into (4):

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i \, B\epsilon_{t-i} = \mu + \sum_{i=0}^{\infty} \Theta_i \, \epsilon_{t-i} \tag{5}$$

Here, the matrices Θ_i are the structural impulse response matrices, and they contain the coefficients required to construct the impulse response function. To illustrate, the equation for the first endogenous variable can be written as:

$$Y_{1t} = \mu_1 + \sum_{k=1}^{K} \theta_{1k,0} \, \epsilon_{k\,t} + \sum_{k=1}^{K} \theta_{1k,1} \, \epsilon_{k\,t-1} + \sum_{k=1}^{K} \theta_{1k,2} \, \epsilon_{k\,t-2} \, + \, ... \tag{6}$$

where $\theta_{jk,i}$ denotes the response of variable j to structural shock k at horizon i, i.e., the (j,k) element of Θ_i.

The structural impulse responses in (6) depend on the model through two distinct channels. First, the moving average coefficients Φ_i are determined by the reduced-form autoregressive coefficients $A_1, ..., A_{p_0}$, so the impulse responses depend on the lag order p_0; in practice, p_0 is unknown and must be estimated, which is the subject of the lag-order selection criteria discussed in Section 3. Second, the mapping from reduced-form to structural form impulse responses requires the identification of B introduced in (2). Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D, the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B. The literature offers several strategies for this purpose, including short-run (recursive) restrictions, often computed by Cholesky decomposition (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Canova and Nicoló, 2002; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. We calibrate our data-generating process to the model of Kilian and Murphy (2014) and accordingly employ a sign-restriction algorithm to identify the impact matrix (though we omit the elasticity bounds for computational feasibility). A key feature of this approach is that sign restrictions deliver only partial identification: rather than a single impact matrix, they identify a *set* of candidate matrices consistent with the a priori restrictions implied by economic theory. This raises several questions about the identification strategy, which are the focus of the next subsection.

2.2 Sign Restrictions

Point identification strategies, such as the Cholesky decomposition, impose an assumption restrictive enough to single out a unique impact matrix; in the Cholesky case, this is a recursive ordering that determines the contemporaneous causal structure among the endogenous variables. Uhlig (2005), however, criticizes such strategies on the grounds that these restrictive assumptions may not reflect the underlying economic reality, but instead may impose the very conclusions the researcher expects to find.

Uhlig (2005) proposes an alternative he terms an "agnostic" identification procedure. The agnostic approach imposes only minimal restrictions—and, crucially, leaves the response of the variable of central interest entirely unrestricted—so that the answer to the research question is determined by the data rather than by the identifying assumptions. The restrictions that are imposed are deliberately weak: they consist only of sign restrictions on a few responses that can be reasonably agreed upon across economists. This minimalism comes at a cost, however. Imposing only sign restrictions does not single out a unique impact matrix; instead, it identifies a set of candidate impact matrices consistent with the assigned signs. This set-identification (or partial identification) strategy is less restrictive, but it comes at the cost of ambiguity, since the object of interest is no longer a single value but a range. Moreover, as emphasized by Uhlig (2005), sign restrictions introduce a further choice that point-identifying schemes do not: the horizon over which the restrictions are imposed. In addition to specifying the sign of a response on impact, the practitioner must decide how many subsequent periods that sign is required to hold—the longer this horizon, the more candidate models are discarded and the narrower the identified set becomes.

This trade-off is described by Manski (2003) as the Law of Decreasing Credibility: the credibility of inference decreases as the strength of the maintained assumptions increases. Point identification strategies impose stronger assumptions and are therefore more informative—they reduce the identified object to a single point—but they are less credible, since their validity rests entirely on assumptions that may not hold. Partial identification strategies, by contrast, are less informative—they deliver a trimmed range rather than a single element—but more credible, precisely because they rely on weaker and more defensible assumptions.

Here, we follow Kilian and Murphy (2014) with slight differences due to computational feasibility.[1] First, we fit VAR(p_0) model to the data of Kilian and Murphy (2014) and obtain the reduced-form autoregressive coefficients ($\hat{A}$) and the variance-covariance matrix of the reduced-form errors ($\hat{\Sigma}$). We then factor this covariance matrix using Cholesky decomposition ($\hat{\Sigma} = PP'$).[2] Next, following Rubio-Ramírez et al. (2010), we generate

[1] Kilian and Murphy (2014) argue that agnostic sign restrictions alone are not sufficient to narrow the set of admissible models adequately, and they therefore impose additional quantitative restrictions; bounds on the implied oil supply and demand elasticities. Additionally, they set dynamic restrictions for a horizon of 12 months. We omit the elasticity restrictions and reduce the dynamic restrictions to 3 months for reasons of computational feasibility: in our finite samples, the acceptance rate under the elasticity bounds is prohibitively low, so that obtaining a sufficient number of admissible models in each of the many Monte Carlo iterations would require a computationally unachievable number of draws.

[2] Kilian and Murphy (2014) uses eigendecomposition here. However, it doesn't make any difference. Any two factors satisfying $BB' = \tilde{B}\tilde{B}' = \hat{\Sigma}$ differ only by an orthogonal rotation ($B = \tilde{B}D$). Since our procedure applies the full set of Haar-distributed rotations to the initial factor, the resulting set of

random orthogonal matrices: we draw a matrix of independent standard normal entries, compute its QR decomposition, and normalize the sign of each column so that the diagonal of (R) is positive. This procedure yields orthogonal matrices (Q) (with $QQ' = I_K$) that are uniformly distributed with respect to the Haar measure, ensuring that the space of admissible rotations is sampled without bias. For each draw, we form a candidate impact matrix $(\tilde{B} = PQ)$ and check whether it satisfies the imposed sign restrictions; candidates that violate the restrictions are discarded. Finally, we retain a select number of admissible impulse response structures and select the median target as our true IRF (population response). For the sake of clarity, we find it appropriate to outline the step-by-step blueprint of this algorithm:

1. Draw a $(K \times K)$ matrix W, where each element is drawn independently from a standard normal distribution: $W_{ij} \sim N(0, 1)$.

2. Compute the QR decomposition $W = QR$, where Q is an orthogonal matrix (the rotation matrix) and R is upper triangular. Normalize the sign of each column of Q so that the diagonal of R is positive, which renders the decomposition unique and ensures that Q is distributed according to the Haar measure.

3. Construct the candidate impact matrix $\tilde{B} = PQ$, and then check whether $\tilde{B}$ satisfies the imposed restrictions.

4. Repeat this process by drawing new W matrices (i.e., obtaining new rotation matrices Q) until the desired number of admissible models is reached.

5. Among the retained models, select the median target m^* as the representative (true) model.

The selection of a representative model is a much-debated issue in the literature, particularly the comparison between the pointwise median and the median target. Since we must obtain a single population response rather than a set—otherwise the mean squared error (MSE) could not be computed against a unique true IRF—we adopt the approach most appropriate for our design. Fry and Pagan (2011) criticize the pointwise median for a compelling reason: the pointwise median does not correspond to any single admissible model, as it combines responses drawn from different models with different impact matrices. Consequently, this synthetic profile may appear plausible but need not correspond to any B satisfying $\hat{\Sigma} = BB'$. This implies that the shocks may not be properly isolated, so the pointwise median cannot be used for variance decomposition. To address this, Fry and Pagan propose the median target approach:

$$m^* = \arg\min_m \sum_{j,k,i} \left(\frac{\theta_{jk,i}^{(m)} - \mathrm{med}_{jk,i}}{\sigma_{jk,i}} \right)^2, \qquad (7)$$
$$\text{where} \quad \mathrm{med}_{jk,i} = \mathrm{median}_m\{\theta_{jk,i}^{(m)}\}, \quad \sigma_{jk,i} = \mathrm{std}_m\{\theta_{jk,i}^{(m)}\}.$$

admissible models is invariant to the initial decomposition. The only element we need is an initial factor to run the algorithm.

Solving this minimization yields the admissible model whose impulse responses are closest to the pointwise median ($m^* \in M$) Since m^* is an actual admissible model rather than a synthetic construct, it satisfies $\hat{\Sigma} = BB'$ (which makes sure that shocks are orthogonalized) and can be used for variance decomposition. We adopt this approach due to the coherence concerns described above and take the median target as our representative true IRF whereby its specific impact matrix $\tilde{B}$ determines the DGP.[3]

Identification is an essential part of recovering the underlying impulse response function. As noted earlier, however, it is not the only factor that matters: the specification of the reduced-form VAR also plays a role. Recall that $\Phi_i = \sum_{j=1}^{i} A_j \Phi_{i-j}$ where $A_j = 0$ for $j > p_0$, so the moving-average coefficients, and hence the impulse responses, depend on the lag order (p_0). The practitioner must therefore choose the lag order carefully. We address this in the next section.

2.3 Information Criteria

The true lag order p_0 is a priori unknown when estimating a reduced-form VAR; it is a matter of specification. Adding too many lags may lead to overfitting, which inflates the variance of the impulse response estimates by requiring the estimation of additional parameters. Underfitting is also problematic, as it introduces bias. The bias-variance trade-off is therefore central to lag-order selection. Traditionally, information criteria are used to address this issue, and their working principle is essentially the same (Lütkepohl, 2005, chap. 4):

$$\mathrm{IC}(p) = \ln|\hat{\Sigma}(p)| + \underbrace{c_T, \frac{K^2 p}{N}}\text{penalty term} \tag{8}$$

Information criteria are formulas that embody the bias-variance trade-off described above. The $\ln|\hat{\Sigma}(p)|$ term is the log-determinant of the reduced-form error covariance matrix, estimated by quasi-maximum likelihood; this term penalizes underfitting, since a model with too few lags leaves $\hat{\Sigma}(p)$ large and thus raises $\mathrm{IC}(p)$. The Penalty weight c_T depends on the specific criterion, and the penalty term increases with the lag order; the second component therefore penalizes overfitting. The lag order is estimated by:

$$\hat{p} = \arg\min_{p} \mathrm{IC}(p), \qquad 1 \leq p \leq \bar{p} \tag{9}$$

Following Ivanov and Kilian (2005), we set the maximum lag order to $\bar{p} = 12$ for our monthly data, corresponding to one year. This choice exceeds the largest true lag order considered $p_0 = 10$, ensuring that all criteria can both identify and overfit relative to the true order, and it makes our results directly comparable to those of Ivanov and Kilian

[3]For a visualization of the IRFs selected for the DGP, see Figure 5 in the Appendix.

(2005).[4]

Although there are different types of IC(p), three are mainly used:

$$\text{AIC}(p) = \ln|\hat{\Sigma}(p)| + \frac{2}{N}, K^2 p \tag{10}$$

The Akaike Information Criterion(AIC) (Akaike, 1974) is the least overfitting-averse of these three criteria. It uses $c_T = 2$ and tends to overfit in practice.[5]

$$\text{BIC}(p) = \ln|\hat{\Sigma}(p)| + \frac{\ln N}{N}, K^2 p \tag{11}$$

The Bayesian Information Criterion (BIC), also called the Schwarz Information Criterion (Schwarz, 1978), penalizes overfitting the most. It uses $c_T = \ln N$ and is a consistent estimator of the true lag order $\left(\operatorname*{plim}_{N\to\infty} \hat{p}^{\text{BIC}} = p_0\right)$.

$$\text{HQC}(p) = \ln|\hat{\Sigma}(p)| + \frac{2\ln\ln N}{N}, K^2 p \tag{12}$$

The Hannan-Quinn Information Criterion (Hannan and Quinn, 1979) is neither as harsh against overfitting as AIC nor as tolerant as BIC. It uses $c_T = 2\ln\ln N$ as its penalty weight and, like BIC, is a consistent estimator of the true lag order.

The three criteria share the same fit term and differ only in the penalty weight c_T, which equals 2 for the AIC, $2\ln\ln N$ for the HQC, and $\ln N$ for the BIC. A heavier penalty makes each additional lag more costly and therefore yields a more parsimonious model. Comparing the weights gives:

$$2; \leq; 2\ln\ln N; \leq; \ln N \tag{13}$$

where the first inequality holds for $N \geq 16$ and the second for all N.[6] Consequently, the implied lag-order estimates satisfy:

$$\hat{p}^{\text{BIC}}; \leq; \hat{p}^{\text{HQC}}; \leq; \hat{p}^{\text{AIC}} \tag{14}$$

so that the BIC is the most parsimonious criterion and the AIC the least, with the

[4] We only include AIC, BIC and HQC as information criteria and omit the Likelihood Ratio tests and the Lagrange Multiplier test for lag selection as these have become practically irrelevant in the last 20 years for macroeconomic time series analysis and were also shown to perform poorly in Ivanov and Kilian (2005)

[5] In the literature, the AIC is generally regarded as prone to overfitting. In the VAR setting, however, this tendency is attenuated: adding one more lag requires estimating an entire $(K \times K)$ coefficient matrix, so the penalty grows by $\frac{2K^2}{N}$ at each step. Because each additional lag is this costly, even the AIC—the most lenient of the three criteria—can underfit in practice.

[6] The weights coincide at $N = e^e \approx 15.2$ (for $2 = 2\ln\ln N$) and never cross for $\ln N$ versus $2\ln\ln N$, since $x - 2\ln x > 0$ for all $x > 0$. The condition $\ln N \geq 2$ (BIC versus AIC) holds for $N \geq 8$. Therefore, $N \geq 16$ is the requirement for (14) to hold.

HQC in between (Ivanov and Kilian, 2005). Since $N \geq 16$ holds across all sample sizes in our study, this ordering applies throughout. As shown in Table 1, our Monte Carlo Simulation confirms this ordering. The penalization of BIC is evidenced to be so strong that the lag order selected, $\hat{p}^{\text{BIC}}$, always stays close to 2, regardless of the sample size T and the true lag order p_0. $\hat{p}^{\text{HQC}}$ lies above $\hat{p}^{\text{BIC}}$ and increases more apparently with T; yetit also does not come close to converging to the true lag order. On the other hand, $\hat{p}^{\text{AIC}}$ is consistently much closer to the true lag order and has already nearly reached p_0 at $T = 600$. Therefore, although AIC is not asymptotically consistent, it is the only IC that can come close to the true lag order for realistic sample sizes.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator T	AIC	BIC	HQC
4	240	2.99	1.95	2.06
	360	3.39	2.00	2.16
	480	3.61	2.01	2.40
	600	3.80	2.02	2.66
10	240	6.54	1.96	2.07
	360	8.53	2.00	2.23
	480	8.96	2.00	2.66
	600	9.16	2.02	4.03

Lütkepohl (1985) compares information criteria based on their ability to recover the true lag order. Ivanov and Kilian (2005), by contrast, object to this approach on the grounds of the bias-variance trade-off discussed above. To see why, it is helpful to consider the decomposition of the mean squared error (MSE) of the estimated median target IRF $\hat{\theta}$ compared to the true DGP IRF θ:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2] \tag{15}$$

which can be reformulated to:

$$MSE(\hat{\theta}) = E\left[\left(\hat{\theta} - E[\hat{\theta}]\right)^2 + \underbrace{2\left(\hat{\theta} - E[\hat{\theta}]\right)\left(E[\hat{\theta}] - \theta\right)}_{=0} + \left(E[\hat{\theta}] - \theta\right)^2\right] \tag{16}$$

and finally, after we distribute the expected value operator:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2 \tag{17}$$

we have derived an equation for the bias-variance tradeoff. In small samples, if underfitting produces a reduction in variance large enough to offset the resulting increase in bias, the MSE falls, and if we take the MSE as a proxy for accuracy, this matters more than a criterion's ability to recover the true lag order. The lag order selected by a criterion that underfits may therefore outperform the true lag order; one of the central remarks of Ivanov and Kilian (2005).

Up to this point, our discussion has been rooted entirely in a frequentist framework. However, as Litterman (1986) posits, a "realistic representation of the current state of economic theory" demands that we account for substantial uncertainty regarding the economy's structure. In the following section, we transition to Bayesian methods, which formally integrate this structural uncertainty into the estimation process and share Litterman (1986)'s view.

2.4 Bayesian Model Averaging (BMA)

Bayesian Model Averaging is an averaging mechanism designed to address the problem of model uncertainty (Hoeting et al., 1999; Steel, 2020). For any inference question, there may be more than one model that fits the data well, and committing to a single one ignores the uncertainty inherent in that choice. In our setting, the candidate models are reduced-form VARs of different lag orders. Bayesian Model Averaging retains all of them and forms a weighted average, with each model weighted by its posterior model probability. The averaged result is preferable for two reasons. First, it accounts for model uncertainty in addition to parameter uncertainty, rather than conditioning on a single model as if it were known to be correct. Second, averaging over all models in proportion to their posterior probabilities yields better average performance, as measured by a logarithmic scoring rule, than any single model (Madigan and Raftery, 1994; Hoeting et al., 1999). In this section, we list four different Bayesian Model Averaging methods that we use as a metric (namely, Uniform prior-OLS, Geometric prior-OLS, Joint Grid, and Geometric Grid) and discuss their properties. To clarify, here is the blueprint of the process that we follow:

1. We estimate every reduced form VAR with lag order specification up to $\bar{p} = 12$ (maximum lag order) using OLS, and different $(p; \lambda_1)$ combinations using the BVAR.

2. Then we calculate the weight parameters of each model (we use AIC/BIC for OLS estimations and the marginal data density (MDD) for the BVAR estimations).

3. We draw different amounts of IRFs based on the weights (Higher weight implies more draws from that specific model) and pool them.

4. We choose the median target IRF as discussed before.

This process raises several questions. How are the model weights calculated? Is averaging genuinely better than selecting a single model? What do the six BMA variants share, and how do they differ? We address these in turn.

We find it important to discuss the common properties of the four different BMA approaches listed above. We follow the canonical treatment of Hoeting et al. (1999) for this part. First, we start with the problem of choosing the right weight parameters (so that we can average the models based on the weights). Let Δ be our parameter of interest (Impulse response functions) and let $M_1, \ldots, M_K$ be our candidate models (different reduced form VAR models). The posterior distribution of Δ given the data y (step 3 defined in the blueprint), averaged over all models, is:

$$p(\Delta \mid y) = \sum_{k=1}^{K} p(\Delta \mid M_k, y) p(M_k \mid y) \tag{18}$$

so this is the weighted average of the posterior distributions conditioned on the model $p(\Delta \mid M_k, y)$, weighted by $p(M_k \mid y)$ (posterior probability of model conditioned on data). By Bayes' theorem applied at the level of models, this weight is:

$$p(M_k \mid y) = \frac{p(y \mid M_k) p(M_k)}{\sum_{l=1}^{K} p(y \mid M_l) p(M_l)} \tag{19}$$

where $p(M_k)$ is the prior probability assigned to model M_k and $p(y \mid M_k)$ is its marginal data density—the likelihood integrated over the model's parameters under their prior:

$$p(y \mid M_k) = \int p(y \mid \theta_k, M_k) p(\theta_k \mid M_k) \, d\theta_k. \tag{20}$$

The marginal data density (MDD or marginal likelihood) in (20) is central to our procedure, as it is the primary input to the model weights in (19).

These 3 equations defined above are the backbones of the Bayesian Model Averaging methodology. In practice, these three equations map directly onto the steps of our procedure. Equation (20)—the marginal data density—is evaluated first: for each candidate model, we obtain a measure of how well it explains the data once its parameters are integrated out. For the BVAR models, indexed by a lag order and tightness pair (p, λ_1), this quantity is the marginal data density returned in closed form by the conjugate Normal-Wishart estimator; for the OLS models, indexed by lag order alone (p), it is approximated by BIC and AIC-[7] Equation (19) then converts these marginal data densities into model weights: each is multiplied by its model prior $(p(M_k))$—uniform

[7] Steel (2020, chapt. 3.7) mentions that marginal data density is approximated by the Bayesian Information Criterion when the closed form of equation (20) does not exist, as BIC asymptotically converges to marginal data density. Also, Burnham and Anderson (2002) justifies the usage of AIC as an approximation of the marginal data density. Therefore, we use both criteria.

or geometric—and normalized so that the weights across all candidate models sum to one (denominator). These weights are the posterior model probabilities. Finally, (18) is realized not by analytical combination but by Monte Carlo: from each model we draw admissible impulse responses in a number proportional to its weight, so that models the data favor contribute more draws, and we collect all draws into a single pool. This pool is the Monte Carlo representation of the model-averaged distribution in (18). From it, we select the median target as our single, coherent estimate of the impulse response, completing the averaging.

So far, we have described the common structure they share. However, we use four different BMA methods, and each has its own priors or estimation methods. We discuss the differences below.

Weights and Priors

We use two weight sources and two model-space priors (uniform prior and geometric prior). Each combination yields a different metric:

	Uniform prior	**Geometric prior ($\theta = 0.5$)**
OLS (BIC-weighted, p)	Uniform-OLS	Geometric-OLS
OLS (AIC-weighted, p)	Uniform-OLS	Geometric-OLS
BVAR (MDD-weighted, p, λ_1)	Joint Grid	Geometric Grid

We find it fruitful to discuss model priors that we use ($p(M_k)$) first. The uniform prior is straightforward. It represents the belief that every candidate model has the same probability of being the true model (reflecting the absence of any prior preference among lag orders):

$$p(M_1) = p(M_2) = \cdots = p(M_K) = \frac{1}{\bar{p}} \tag{21}$$

On the other hand, geometric prior constructs the prior as:

$$p(M_k) \propto \theta^p \quad \text{where} \quad \theta = 0.5 \tag{22}$$

We set $\theta = 0.5$, under which each additional lag halves the prior weight—equivalently, the prior demands that the marginal data density favor a longer model by at least a factor of two before the additional lag is preferred. This value is a deliberate, interpretable benchmark rather than an optimized choice: it imposes a moderate preference for parsimony, neither so weak as to coincide with the uniform prior nor so strong as to dominate the evidence from the marginal data density.

For models estimated via OLS, (20) is undefined because there are no priors $p(\theta_k \mid M_k)$; under the frequentist approach, each estimated parameter consumes one full degree of freedom with no shrinkage. We therefore approximate the marginal data density by

$\exp\left(-\frac{1}{2}\mathrm{IC}(p)\right)$, where IC is either BIC or AIC. As derived by Schwarz (1978) and Raftery (1995), this serves as an asymptotic approximation to the log marginal data density that, in large samples, does not depend on a prior. For the BVAR models, by contrast, the Normal-Wishart prior—a normal prior on $\boldsymbol{\alpha}$ and an Inverse-Wishart prior on Σ, as introduced below—supplies the joint density $p(\theta_k \mid M_k)$ that (20) integrates over.[8] Because this conjugate structure makes the integral available in closed form, the marginal data density $m(y \mid p, \lambda_1)$ is computed directly.

By using (19) we calculate the weights for the models estimated via OLS:

$$w_{\mathrm{OLS}}(p) = \frac{\exp\left(-\frac{1}{2}\mathrm{IC}(p)\right)\pi(p)}{\sum_{p'=1}^{\bar{p}} \exp\left(-\frac{1}{2}\mathrm{IC}(p')\right)\pi(p')} \tag{23}$$

where $\pi(p) = \frac{1}{\bar{p}}$ under the uniform prior and $\pi(p) \propto \theta^p(\theta = 0.5)$ under the geometric prior. For the BVAR models, indexed by the pair (p, λ_1), the weight is:

$$w_{\mathrm{BVAR}}(p, \lambda_1) = \frac{m(y \mid p, \lambda_1)\pi(p)}{\sum_{p'=1}^{\bar{p}} \sum_{\lambda_1'} m(y \mid p', \lambda_1')\pi(p')} \tag{24}$$

where $m(y \mid p, \lambda_1)$ is the calculated marginal data density (20) with joint prior and $\pi(p)$ is the same priors that are described above.

The BVAR models are indexed by pairs (p, λ_1), with the lag order ranging over $p = 1, \ldots, \bar{p} = 12$ and the tightness over the discrete grid $\lambda_1 \in \{0.20, 0.40, 0.60, 0.80\}$.[9] For every pair in this grid, we estimate the Normal-Wishart BVAR and compute its marginal data density. The pairs are then weighted by their posterior model probability (24); pairs whose normalized weight falls below a threshold of 0.01 are discarded, and the weights of the retained pairs are renormalized to sum to one. From each retained pair, we draw impulse responses in proportion to its weight, pool the draws, and select the median target.

Advantages of the Bayesian Model Averaging

First of all, Bayesian Model Averaging explicitly addresses model uncertainty directly. Following Hoeting et al. (1999):

$$\mathrm{Var}[\Delta \mid y] = \sum_k \left(\mathrm{Var}[\Delta \mid y, M_k] + \hat{\Delta}_k^2\right) p(M_k \mid y) - E[\Delta \mid y]^2 \tag{25}$$

[8] "For an introduction to the BVAR methodology, see Section D of the Appendix; for a thorough treatment, see Aydın (2026).

[9] This set is relatively small but captures both tighter and looser hyperparameters. We have to limit it due to computational limitations.

Here, $\hat{\Delta}_k = E[\Delta \mid y, M_k]$. After decomposing $\mathrm{Var}[\Delta \mid y]$, it is possible to see that BMA framework accounts for the model uncertainty:[10]

$$\mathrm{Var}[\Delta \mid y] = \underbrace{\sum_k \mathrm{Var}[\Delta \mid y, M_k] p(M_k \mid y)}_{\text{within-model}} + \underbrace{\sum_k \left(\hat{\Delta}_k - E[\Delta \mid y]\right)^2 p(M_k \mid y)}_{\text{across-model}} \tag{26}$$

From (26) we see that if a single model dominated the posterior—such that $\hat{\Delta}_k = E[\Delta \mid y]$ for that model—the across-model term would vanish and $\mathrm{Var}[\Delta \mid y]$ would reduce to the within-model variance alone. That is clearly an advantage of Bayesian Model Averaging since it allows the practitioner to estimate the posterior variance in a more accurate way.

Second, model averaging has a formal optimality property: Madigan and Raftery (1994) show that, under a logarithmic scoring rule, the model-averaged predictive distribution outperforms any single model. While our study concerns estimation rather than forecasting, this result underpins the broader case for averaging. Hoeting et al. (1999) provide a concrete illustration in which even the highest-probability model carries only 17% of the posterior model probability, underscoring that model uncertainty is often substantial. Therefore, it makes sense to use Bayesian Model Averaging in order to account for model uncertainty. Our central motivation for using this approach is to assess whether accounting for model uncertainty can also ensure more accurate IRF estimation.

Having established the theoretical framework and estimation methodology, we now turn to the simulation design to evaluate the empirical performance of the competing approaches.

3 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series, and additionally, we do not have additional robustness by including several different series with different macroeconomic indicators and movements. The reason for this limitation is our tight time constraint. We use the dataset from Kilian and Murphy (2014) containing the following 4 variables: percent change in global oil production, real activity index from Kilian (2009), the log real price of oil, and changes in OECD crude oil inventories. Just as in the original study, we first estimate a VAR with a set lag order of these 4 variables and then use the sign restrictions approach explained earlier to identify structural IRFs and select the median target.

[10] For the derivation of this decomposition refer to Section E in the Appendix.

The DGP is formed by structural VAR model using the structural impact matrix $\tilde{B}$ that corresponds to the median target IRF as the parameter for the structural shocks e_t, which are drawn from an i.i.d. standard normal distribution. The formula for the DGP is represented by a structural VAR model using the estimated reduced form parameters A (for the lag coefficients y_{t-p}) and V (for the monthly dummy parameters d_t) and the structural impact matrix $\tilde{B}$ as follows:

$$y_t = V d_t + A y_{t-p} + \tilde{B} e_t \tag{27}$$

We estimate these parameters using the original data for two different lag orders, $p_0 \in \{4, 10\}$ and create DGPs with different sample sizes, $T \in \{240, 360, 480, 600\}$, where $T = N - \bar{p}$. $\bar{p}$ is the set of lag orders that are considered for both the IC and the BVAR estimations; it is uniformly set to $\bar{p} = 12$ to ensure that the true lag order p_0 is part of the set. We restrict the lag order selection p_0 to only include one high and one low lag order and drop $T = 300$ in comparison to the setup of Ivanov and Kilian (2005) in order to save computational power for the sign restriction decomposition. For sampling, we use random draws of length p_0 from the original dataset as initial values, then apply the parametric bootstrap using the SVAR described above and generate sequences of lengths T.

On each of these samples, VARs fitted by lag orders selected by both the three ICs as well as the six different BMA methods are estimated, a fixed number of admissible IRFs found, and the median target IRF determined.

Finally, we estimate the key performance parameter, the relative MSE, by calculating the geometric mean over the ratio of the squared errors of the median target IRFs $\hat{\theta}_j$ computed by method j in relation to the squared errors of the median target IRF by a model fit with the true lag order $\hat{\theta}_{p_0 i}$ over all Monte Carlo iterations from $1, ..., R$:

$$MSE_j^{relative} = \exp\left(\frac{1}{n}\sum_{i=1}^{R}\ln(\frac{SE_{ji}}{SE_{p_0 i}})\right) \tag{28}$$

The geometric mean is preferred over the arithmetic mean as the ratios are bounded on the left side at 0, but unbounded on the right side. Outliers on the right side are hence overweighed in the arithmetic mean. Normalization by the OLS VAR estimation with p_0 ensures that the performance in each iteration is weighed by how well a correctly identified model performs.

For the computation of the DGP parameters, we set the number of selected admissible IRFs to 1 million in order to get a highly accurate representation of the real world movements. However, during the Monte Carlo simulation we have to reduce that number to 1,000, which is seen as the minimum requirement for accurate set identification, albeit still with a larger variance. We set the number of Monte Carlo iterations to $R = 500$.

We understand the concern that this may be too little for eliminating all Monte Carlo biases. Yet we have to rely on such a small number as it is the absolute maximum that our hardware allows for, even though we have optimized the code by employing parallel processing and delegation to C-based processes using the `numba` and `joblib` packages in `python`. All in all, this simulation took 18 hours to complete with 6.5 million admissible IRFs drawn.[11]

4 Results

First, we will look at the main comparison of performance between the different ICs and BMA specifications in terms of recovering an accurate median target IRF. Table 2 displays the full set of relative MSEs. It reveals two main findings:

1. We confirm the results from Ivanov and Kilian (2005) about the performance of the different information criteria. AIC consistently outperforms both BIC and HQC, revealing that the strict penalization of the latter two towards low lag orders is biasing the IRF too much. On the other hand, AIC's milder lag penalization ensures a closer fit to the true lag order p_0 for finite samples, thereby reducing bias. The reduction of the variance by selecting a p much lower than the true lag order as done by BIC and HQC, see Table 1, is not evidenced to reduce the coefficient variance enough to compensate for the additional bias.

2. Hedging the risk of lag selection by employing BMA is the best option in nearly all scenarios. For smaller sample sizes, $T = 240$ and $T = 360$, joint grid BMA is dominating all other methods by a solid margin. The OLS fitted BMA using AIC as the criterion is the best criterion in two scenarios, once without and once with geometric decay. However, for $p_0 = 10$ and $T = 600$, joint grid BMA's relative MSE is only slightly higher. Only for the highest sample size $T = 600$ and the lower true lag order $p_0 = 4$ does the conventional VAR fitted by an AIC-selected lag order perform the best. This may be explained by the finding that $\hat{p}^{AIC}$ has basically converged to p_0 at $T = 600$ (See Table 1) while the sample size is already large enough in order to accurately determine the coefficients for four lags.

Table 3 makes the comparison to AIC for all different scenarios clearer by setting MSE_{AIC} as the benchmark. Both OLS BMA with AIC as well as joint grid BMA outperform regular AIC in the majority of scenarios. Meanwhile, all other versions of BMA are actually not consistently better than simple AIC. The sensitivity of the BMA hedging advantage to the exact specification requires further examination.

To understand this, we have to consider Figures 1 and 2 that present the weighting distribution of the different BMA methods for $p_0 = 4$. The reason for the BIC-weighted

[11] The computation included both the methods used in this paper as well as in Aydın (2026).

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging.

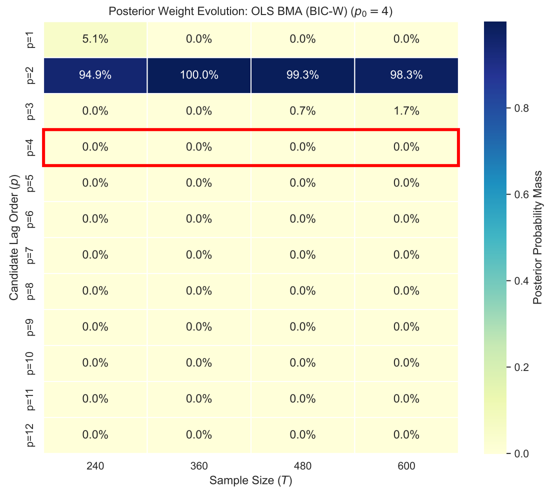
p_0	T	AIC	BIC	HQC	OLS BMA (BIC)	OLS BMA (Geom. BIC)	OLS BMA (AIC)	OLS BMA (Geom. AIC)	Joint Grid BMA	Geom. Grid BMA
4	240	1.083	1.108	1.129	1.100	1.163	1.082	1.175	**0.900**	1.127
	360	1.117	1.274	1.296	1.142	1.235	1.035	1.138	**1.026**	1.046
	480	0.986	1.133	1.124	1.148	1.150	1.031	0.996	**0.898**	1.130
	600	**0.989**	1.145	1.118	1.218	1.129	1.032	1.124	1.113	1.109
10	240	1.033	1.018	1.021	1.010	0.842	0.979	0.982	**0.800**	0.849
	360	0.811	0.904	0.878	1.163	1.057	0.893	0.884	**0.787**	0.993
	480	0.989	1.190	1.185	1.052	1.283	**0.883**	1.210	1.019	0.898
	600	1.036	1.260	1.259	1.293	1.100	0.883	**0.875**	0.878	0.964

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.

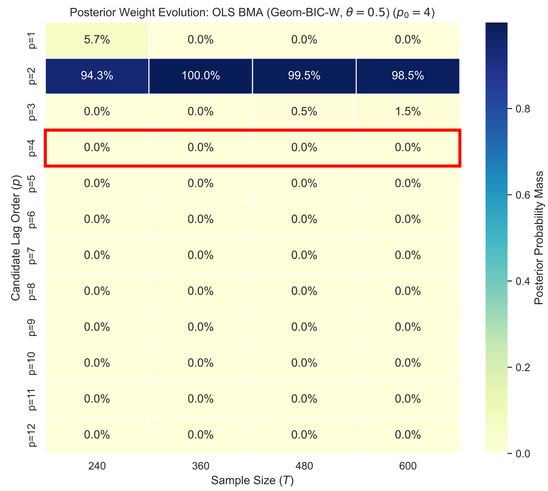
Table 3: Relative Mean Squared Error of Bayesian Model Averaging Specifications vs. AIC Baseline.

p_0	T	OLS BMA (BIC)	OLS BMA (Geom. BIC)	OLS BMA (AIC)	OLS BMA (Geom. AIC)	Joint Grid BMA	Geom. Grid BMA
4	240	1.015	1.074	0.999	1.085	**0.831**	1.040
	360	1.023	1.106	0.927	1.019	**0.919**	0.937
	480	1.164	1.166	1.046	1.010	**0.911**	1.147
	600	1.231	1.141	**1.043**	1.136	1.125	1.122
10	240	0.978	0.815	0.948	0.950	**0.775**	0.821
	360	1.434	1.304	1.102	1.090	**0.970**	1.224
	480	1.064	1.297	**0.893**	1.224	1.031	0.908
	600	1.248	1.062	0.852	**0.845**	0.847	0.930

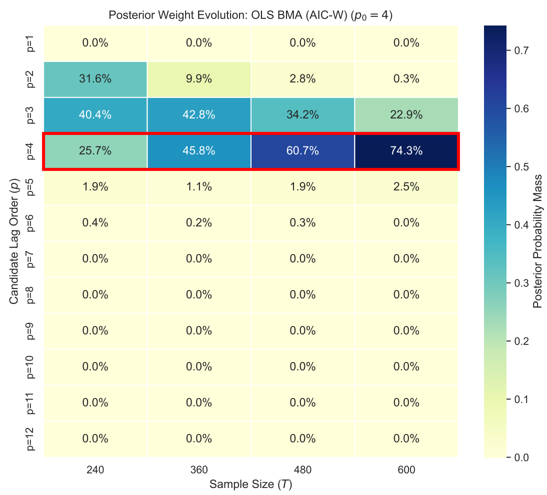
Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.



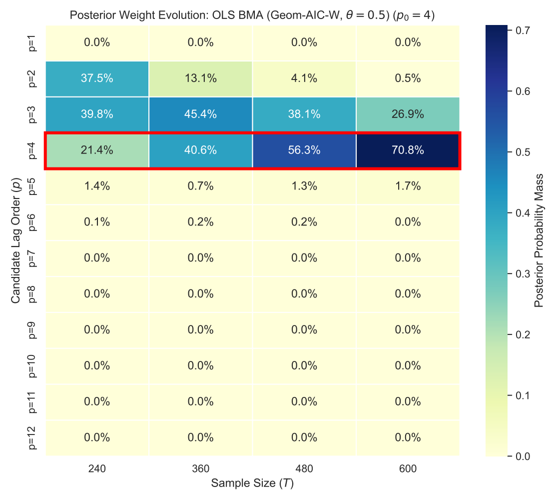
(a) OLS BMA (BIC-weighted)



(b) OLS BMA (Geom. BIC-weighted, $\theta = 0.5$)

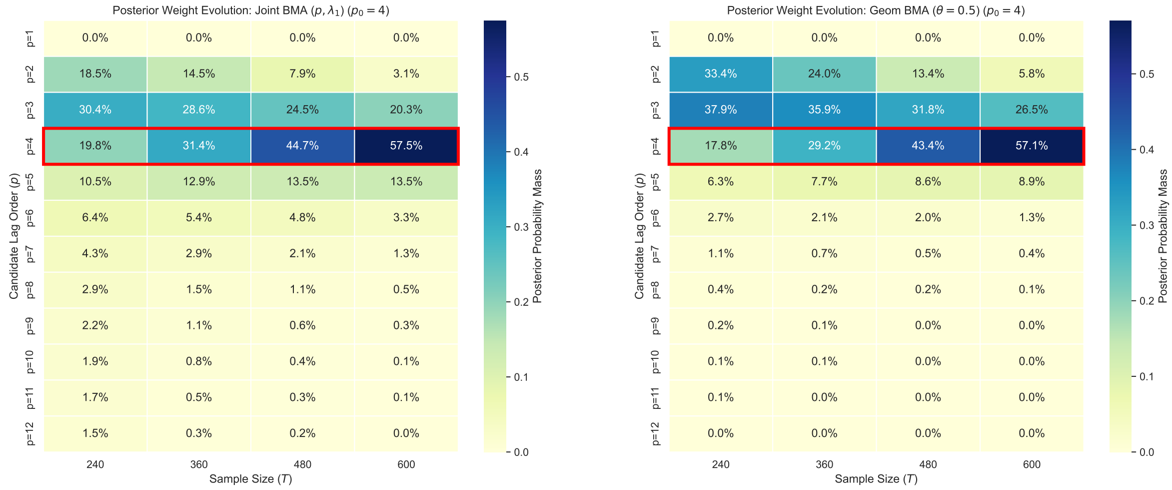


(c) OLS BMA (AIC-weighted)



(d) OLS BMA (Geom. AIC-weighted, $\theta = 0.5$)

Figure 1: Posterior Weight Evolution across the four OLS BMA architectures ($p_0 = 4$).



(a) Joint Grid BMA

(b) Geometric Grid BMA

Figure 2: Posterior Weight Evolution for the two BVAR-based BMAs ($p_0 = 4$).

OLS BMA to fail is very apparent. The strong penalty of BIC forces it to rely very heavily on the lag order $p = 2$ in finite samples, thereby basically eliminating the diversification of BMA. As BIC generally doesn't produce models with good IRF inference, this also applies to these BMAs.

Once AIC is introduced to OLS BMA, the goal of BMA—spreading out the estimation on a selection of lag orders that are determined to be probable—is fulfilled. AIC's weaker penalization of higher lag orders ensures that four to five different ps are assigned positive weights. The weighting becomes more concentrated on the true lag order $p_0 = 4$ as T increases. The introduction of the geometric decay parameter $\theta = 0.5$ distorts the weighting towards lower lag orders. This moves the estimation further away from the correctly fitted model, resulting in a larger underfitting bias. Table 3 shows that the reduction in parameter variance does not compensate for this bias.

The joint grid approach provides the most widely distributed weighting scheme. Figure 2 aggregates the weights from the grid at the lag order p-level, i.e., weights assigned to the combination of an individual p with different λ_1-values are added up. For lower sample sizes, nearly all lag orders are considered in the BVAR estimated BMAs. Even for the largest $T = 600$, the highest weight is just above 50%. Hence, a wider diversification, also reaching into the territory of overfitting lag orders $p > p_0$, can improve the accuracy of the determined median target IRF when combined with BVAR estimation. The inclusion of higher lag orders at first seems counterproductive. Yet, due to the additional shrinkage of parameters introduced by BVAR, it can reduce the risk of lag order uncertainty further while maintaining accurate parameter estimation for the crucial lower lags. The combined optimization of weighting on lag order p as well as the hyperparameter λ_1 ensures that this shrinkage is amplified for those models that use overparameterized specifications. Figure 3 illustrates how, as an example for the scenario of $T = 600$ and

$p_0 = 4$, the weights among different combinations of p and λ_1 are towards the lower end of $\lambda 1$ as p increases above p_0. Although MDD optimizes on Out-of-Sample performance and therefore may select lower than optimal tau-values for IRF accuracy (Paper A), it appears to work well for joint grid BMA estimation for inference of accurate IRFs. The shift towards lower lag orders introduced by the geometric decay parameter θ, on the other hand, cannot be evidenced to be of advantage for a BMA model estimated by the grid BVAR approach.

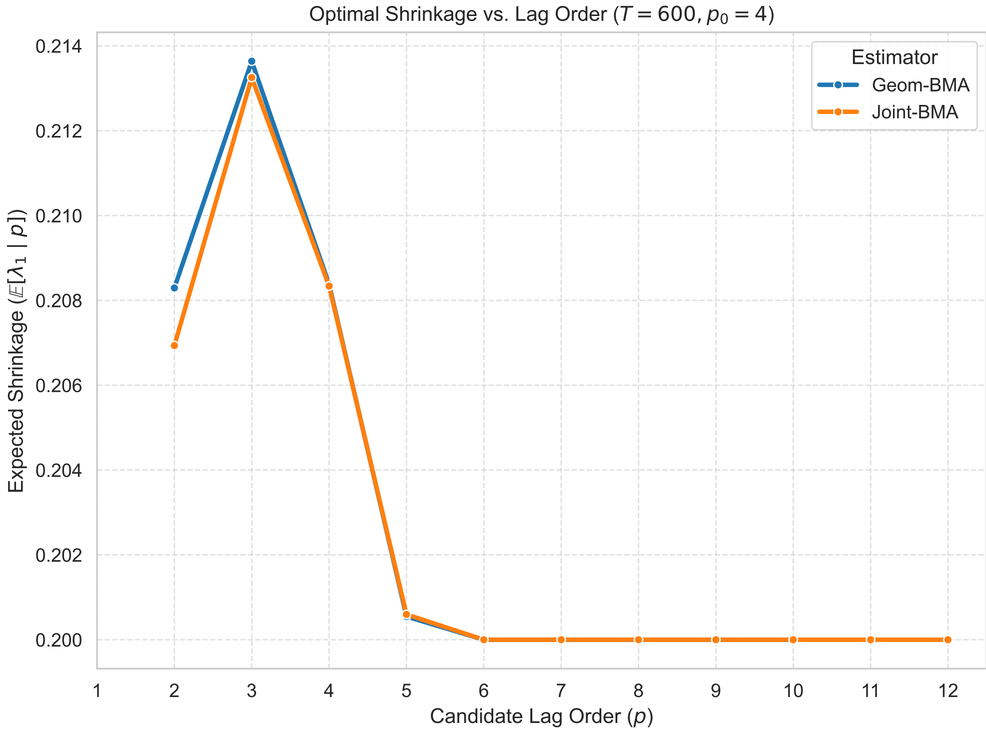
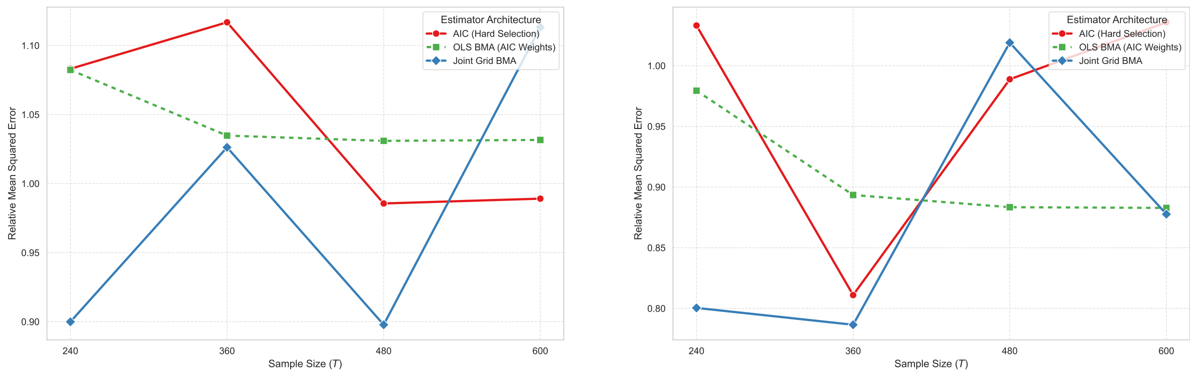


Figure 3: The expected weight of λ_1 given p changes with p ($p_0 = 4$).

A caveat for the interpretation of these results, however, remains: the low number of Monte Carlo iterations and admissible IRFs drawn per model. We cannot exclude that some part of the variation in relative MSEs may be driven by pure sampling variation. For example, the general increase in relative MSEs across all methods at $T = 360$ and $p_0 = 4$ can barely be explained by the theoretical properties. If you look at how the methods perform over different sample sizes T (Figure 4), you can also not deduce a reasonable pattern. Only the simple AIC estimation has a downward movement that basically converges to the lowest possible MSE at $T = 360$. However, both BMA methods shown, AIC OLS as well as joint grid, do not show a clear trend. We suspect that this is due to the complexity of these methods in each iteration whose number may not be enough to eliminate Monte Carlo biases.



(a) $p_0 = 4$ (b) $p_0 = 10$

Figure 4: Performance of BMA vs Hard Selection at different sample sizes T.

5 Conclusion

This paper revisits the fundamental challenge of lag order specification in Structural Vector Autoregressions, updating the foundational Monte Carlo analysis of Ivanov and Kilian (2005) to reflect modern empirical practices. Specifically, we evaluate the performance of traditional Information Criteria (IC) against modern Bayesian Model Averaging (BMA) techniques in recovering the true median target Impulse Response Function (IRF) within a set-identified framework using sign restrictions. Utilizing a realistic data-generating process based on the global oil market model of Kilian and Murphy (2014), we analyze the bias-variance trade-offs inherent in single-model selection versus model averaging across various finite sample sizes.

Our findings underscore two major practical implications for applied macroeconomists. First, when relying on single-model selection, the Akaike Information Criterion (AIC) remains the superior choice for IRF inference. Strict criteria such as the BIC and HQC penalize parameterization too heavily in finite samples, leading to a severe underfitting bias that heavily distorts structural dynamics. AIC's milder penalization yields models that converge closer to the true lag order, confirming that avoiding underfitting bias is paramount for accurate IRF recovery, even if it comes at the cost of slightly higher parameter variance.

Second, and more importantly, our results demonstrate that hedging lag selection risk through Bayesian Model Averaging consistently outperforms single-model selection. In particular, the joint-grid BMA approach, which averages across both lag orders (p) and BVAR shrinkage hyperparameters (λ_1), emerges as the most robust methodology across the majority of sample sizes and true lag structures. This approach effectively mitigates model uncertainty by distributing weights across a broad spectrum of plausible lag orders. Crucially, it manages the risk of overfitting by dynamically assigning tighter shrinkage to overparameterized specifications ($p > p_0$), thereby preserving the precision of critical lower-lag estimates while still capturing higher-order dynamics. Conversely, introducing a geometric prior to force parsimony proved counterproductive, as it artificially shifted weight toward lower lags and reintroduced the very underfitting bias that BMA seeks to avoid. Only in scenarios with very large sample sizes ($T = 600$) and a low true lag order ($p_0 = 4$) does a simple AIC-selected OLS model rival the joint-grid BMA, as parameter variance naturally decreases.

We acknowledge that the severe computational intensity of set identification via sign restrictions imposes natural limitations on our simulation design, restricting the number of Monte Carlo iterations to $R = 500$ and admissible draws to 1,000 per iteration. While this introduces some sampling variation, the persistent outperformance of the joint-grid approach provides a compelling baseline. Future research could expand upon this framework by utilizing more efficient sampling algorithms or greater computational resources to confirm these findings across a wider array of data-generating processes and variable dimensions.

Ultimately, this study advocates for a shift in standard empirical practice. Rather than staking IRF inference on a single, potentially misspecified lag order determined by an information criterion, researchers are well-advised to embrace Bayesian Model Averaging. By jointly averaging over lag structures and dynamically scaling parameter shrinkage, practitioners can significantly improve the credibility, robustness, and accuracy of their structural impulse response estimates.

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Appendix

A DGP IRFs

Figure 5 visualizes the IRFs selected for the DGP of $p_0 = 4$ and $p_0 = 10$. You can see that they exhibit the same general pattern but that a higher lag order captures more complex movements.

B Derivation of the Relationship Between Structural Form Innovation Terms and Reduced Form Errors

For clarity, several intermediate steps were omitted in the main text. We collect them here. To derive the relationship in equation (2), we begin from the structural form:

$$Y_t = B_0 Y_t + B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \tag{29}$$

Moving the contemporaneous term $B_0 Y_t$ to the left-hand side gives:

$$\underbrace{(I - B_0)}_{B^*} Y_t = B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \tag{30}$$

Premultiplying by the inverse of B^* yields the reduced form:

$$Y_t = \underbrace{(B^*)^{-1} B_1}_{A_1} Y_{t-1} + \underbrace{(B^*)^{-1} B_2}_{A_2} Y_{t-2} + \cdots + \underbrace{(B^*)^{-1} B_{p_0}}_{A_{p_0}} Y_{t-p_0} + \underbrace{(B^*)^{-1} \varepsilon_t}_{e_t} \tag{31}$$

which is the explicit reduced-form representation. The relationship defined in (2) is now apparent: writing $(B^*)^{-1} = B$, we have:

$$e_t = B\varepsilon_t \tag{32}$$

Here, B_0 and B denote distinct objects: B_0 is the structural contemporaneous coef-

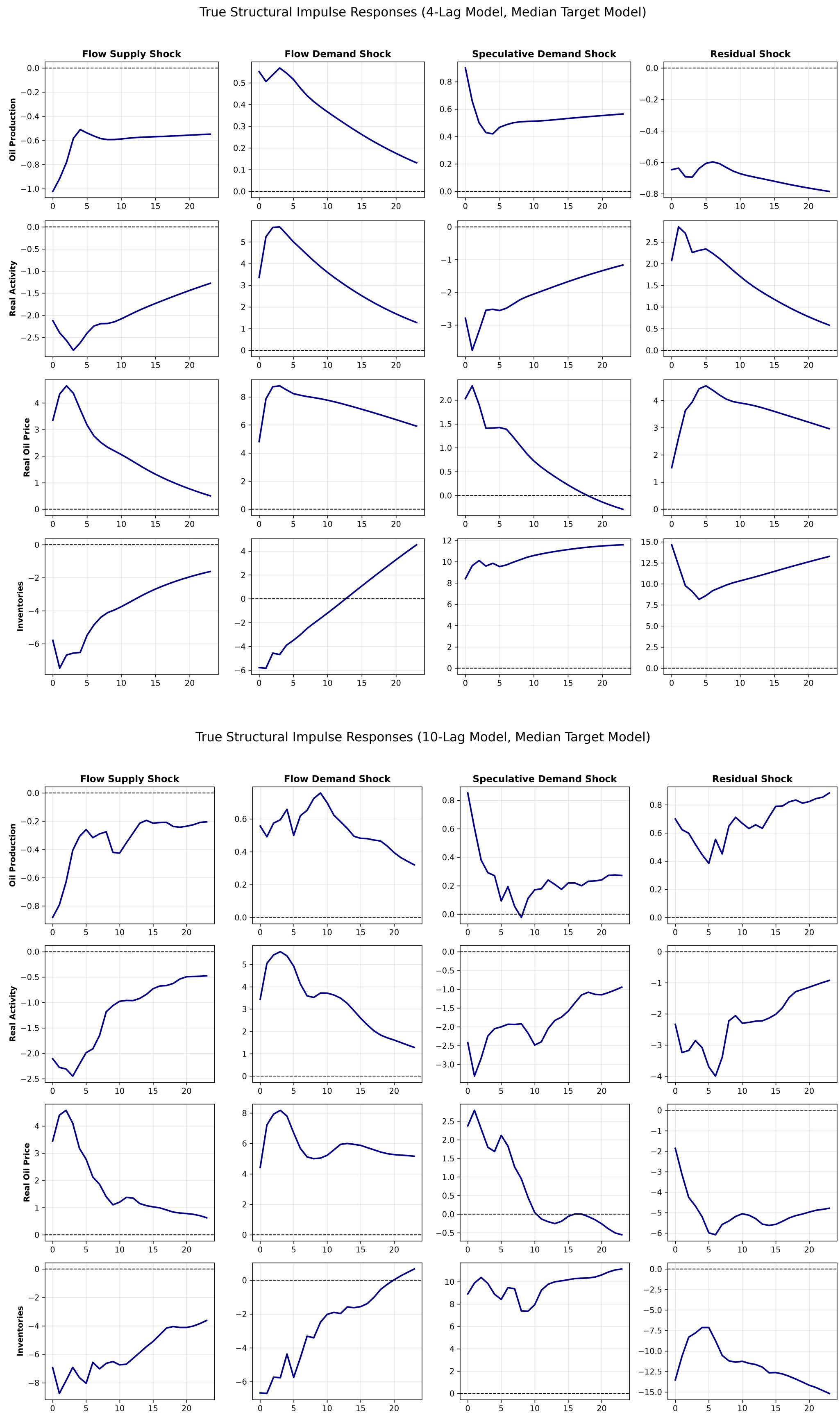


Figure 5: DGP IRFs for $p_0 = 4$ and $p_0 = 10$

ficient matrix, which governs the simultaneous interactions among the endogenous variables in the structural form, whereas $B = (I - B_0)^{-1}$ is the impact matrix that maps the unit-variance structural shocks ε_t into the reduced-form innovations e_t. The former is a primitive of the structural model; the latter is the object recovered through the identification procedure in the main text.

C Derivation of the Recursive Coefficients and the Mean

We now derive the recursion in (4). It is convenient to define the autoregressive lag polynomial:

$$A(L) = I - A_1 L - A_2 L^2 - \cdots - A_{p_0} L^{p_0} \tag{33}$$

Its inverse admits a power-series representation in the lag operator:

$$A(L)^{-1} = \Phi(L) = \sum_{i=0}^{\infty} \Phi_i L^i = \Phi_0 + \Phi_1 L + \Phi_2 L^2 + \cdots \tag{34}$$

Which leads to:

$$A(L)A(L)^{-1} = A(L)\Phi(L) = (I - A_1 L - A_2 L^2 - \cdots)(\Phi_0 + \Phi_1 L + \Phi_2 L^2 + \cdots) = I \tag{35}$$

Matching coefficients on equal powers of L gives:

$$\begin{aligned} I\Phi_0 = I_K &\implies \Phi_0 = I_K \\ \Phi_1 - A_1\Phi_0 = 0 &\implies \Phi_1 = A_1\Phi_0 \\ \Phi_2 - A_1\Phi_1 - A_2\Phi_0 = 0 &\implies \Phi_2 = A_1\Phi_1 + A_2\Phi_0 \end{aligned}$$

and, in general, the recursion:

$$\Phi_i = \sum_{j=1}^{i} A_j \Phi_{i-j} \tag{36}$$

The unconditional mean is $\mu = (I - A_1 - A_2 - \cdots - A_{p_0})^{-1} v$ provided the process is weakly stationary. To derive it, take expectations of (1) under weak stationarity, so that $\mathbb{E}[Y_t] = \mu$ for all t:

$$\mathbb{E}[Y_t] = v + A_1\mathbb{E}[Y_{t-1}] + A_2\mathbb{E}[Y_{t-2}] + \cdots + A_{p_0}\mathbb{E}[Y_{t-p_0}] \iff [I - A_1 - A_2 - \cdots - A_{p_0}]\mu = v \tag{37}$$

which gives:

$$\mu = [I - A_1 - A_2 - \cdots - A_{p_0}]^{-1} v \tag{38}$$

D Bayesian VAR (BVAR)

The frequentist lag-order selection problem and Bayesian shrinkage differ fundamentally. The frequentist approach suffers from the curse of dimensionality: each additional lag adds a full block of coefficients to the system, and every estimated parameter consumes one degree of freedom. The Bayesian approach reframes this problem. The estimated value of each parameter is determined jointly by the prior and the data, through the posterior. The posterior mean can be understood as a weighted average:

$$\text{posterior} \approx w(\text{data estimate}) + (1 - w)(\text{prior mean}) \tag{39}$$

where the weight w reflects the tightness of the prior. Because part of each coefficient is supplied by the prior rather than estimated freely from the data, no parameter consumes a full degree of freedom. The curse of dimensionality is thus addressed not by discarding lags, as selection does, but by disciplining their coefficients (Lütkepohl, 2005, chapt. 5).

To evaluate the BVAR models in our averaging grid, we must specify the likelihood and prior. Assuming a Gaussian VAR, the likelihood function is normally distributed. To shrink the coefficients $\boldsymbol{\alpha}$ toward our prior belief (a white noise process, setting the prior mean $\boldsymbol{\alpha}^* = 0$), we utilize the Minnesota prior framework to govern the prior variance. The central hyperparameter is λ_1, the overall tightness, which dictates the strength of the shrinkage. High λ_1 lets the data dominate, while small λ_1 prioritzes the prior belief. The variance of distant lags is shrunk further by a decay factor governed by λ_3, and exogenous/deterministic terms are controlled by λ_4.

However, the standard Minnesota prior treats the residual covariance matrix Σ as fixed. Because our BMA approach requires estimating Σ alongside the coefficients while maintaining an analytically tractable marginal data density—the integral in (20)—we employ the conjugate Normal-Wishart prior:

$$\Sigma \sim \mathcal{IW}(S_0, \alpha_0), \quad \boldsymbol{\alpha} \mid \Sigma \sim \mathcal{N}(\boldsymbol{\alpha}^*, \Sigma \otimes \Phi_0) \tag{40}$$

Here, Σ is assigned an Inverse-Wishart prior, and the coefficients $\boldsymbol{\alpha}$, conditional on Σ, retain a normal prior. The matrix Φ_0 dictates the variance structure based on our hyperparameters $(\lambda_1, \lambda_3, \lambda_4)$.

This conjugate structure—an Inverse-Wishart prior on Σ combined with a conditionally normal prior on $\boldsymbol{\alpha}$—keeps the posterior in the Normal-Inverse-Wishart family. Crucially for our methodology, this ensures the marginal data density $m(y \mid p, \lambda_1)$ is available in closed form, allowing us to compute exact posterior model probabilities. The cost of this analytical tractability is that the prior covariance must take a specific Kronecker structure $(\Sigma \otimes \Phi_0)$. Because Φ_0 is shared across all equations, it forces the cross-variable tightness hyperparameter to $\lambda_2 = 1$, meaning imposing looser or tighter shrinkage on cross-variable coefficients compared to own-lag coefficients is not possible. By iterating over different combinations of (p, λ_1) in our grid, we rely on this closed-form density to calculate the exact weights used for Bayesian Model Averaging.

E Bayesian Model Averaging Variance Decomposition

In Bayesian Model Averaging (BMA), the total variance of a quantity of interest Δ given the data y can be decomposed into two components: the within-model variance and the across-model variance.

Equation (25) gives us:

$$\mathrm{Var}[\Delta \mid y] = \sum_k \left(\mathrm{Var}[\Delta \mid y, M_k] + \hat{\Delta}_k^2\right) p(M_k \mid y) - E[\Delta \mid y]^2 \tag{41}$$

where, $\hat{\Delta}_k = E[\Delta \mid y, M_k]$. Defining $V_k := \mathrm{Var}[\Delta \mid y, M_k]$, $w_k := p(M_k \mid y)$ and $m := E[\Delta \mid y] = \sum_{k=1}^K \hat{\Delta}_k w_k$, equation (41) becomes:

$$\mathrm{Var}[\Delta \mid y] = \sum_{k=1}^K (V_k + \hat{\Delta}_k^2) w_k - m^2 \tag{42}$$

After splitting the sum:

$$\mathrm{Var}[\Delta \mid y] = \underbrace{\sum_{k=1}^K V_k w_k}_{(I)} + \underbrace{\sum_{k=1}^K \hat{\Delta}_k^2 w_k - m^2}_{(II)} \tag{43}$$

Consider the quantity we claim term (II) equals, namely the weighted dispersion of the model means around the averaged mean:

$$\sum_k (\hat{\Delta}_k - m)^2 w_k \tag{44}$$

After expanding the square:

$$\sum_k (\hat{\Delta}_k - m)^2 w_k = \sum_k \hat{\Delta}_k^2 w_k - 2m \sum_k \hat{\Delta}_k w_k + m^2 \sum_k w_k \tag{45}$$

We know that:

$$\sum_{k=1}^K \hat{\Delta}_k \, w_k = m \tag{46}$$

(definition of the averaged mean) and $\sum_{k=1}^{K} w_k = 1$. Substituting these two identities:

$$\sum_k (\hat{\Delta}_k - m)^2 w_k = \sum_k \hat{\Delta}_k^2 w_k - 2m^2 + m^2 = \hat{\Delta}_k^2 w_k - m^2 \tag{47}$$

The right-hand side equals term (II). Therefore the second component of the posterior variance equals (44), the across-model variance. Finally, we can express (43) in closed form:

$$\mathrm{Var}[\Delta \mid y] = \underbrace{\sum_{k=1}^{K} \mathrm{Var}[\Delta \mid y, M_k] p(M_k \mid y)}_{\text{within-model variance}} + \underbrace{\sum_{k=1}^{K} \left(\hat{\Delta}_k - E[\Delta \mid y]\right)^2 p(M_k \mid y)}_{\text{across-model variance}} \tag{48}$$

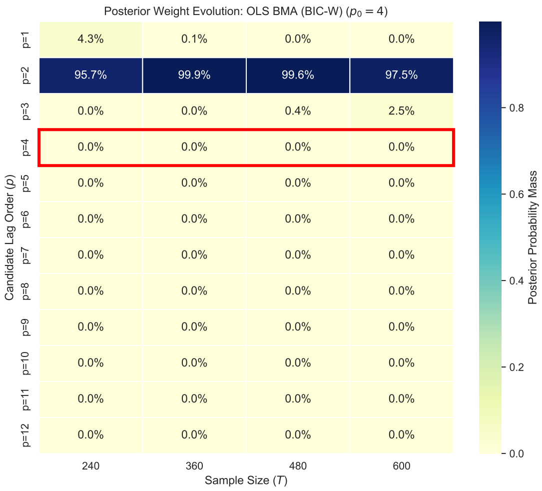
F Posterior Weight Evolution for the BMAs ($p_0 = 10$).

Figures 6 and 7 depict the weighting scheme across sample sizes T for all six different BMA methods. Figure 8 depicts how the expected weight of λ_1 given p changes with p ($p_0 = 10$)

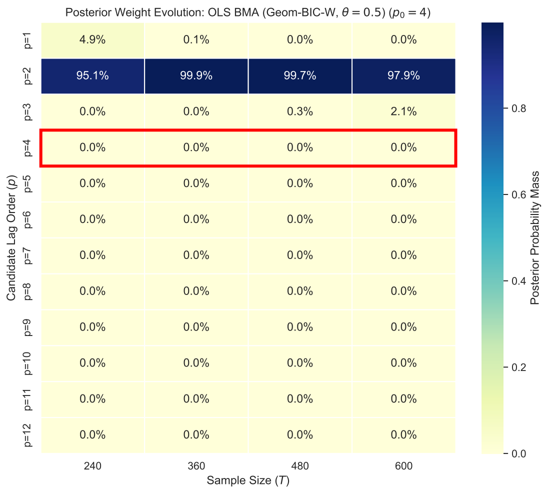
Declaration of AI Usage

During the preparation of this seminar paper, I, Ferdinand Loser, used Claude, Gemini, and ChatGPT for help with coding, research, and adapting writing to academic standards. I have reviewed and edited the content as needed and take full responsibility for the content.

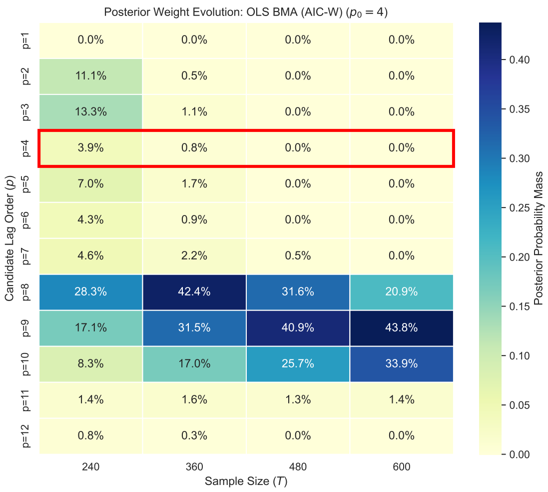
Ferdinand Loser



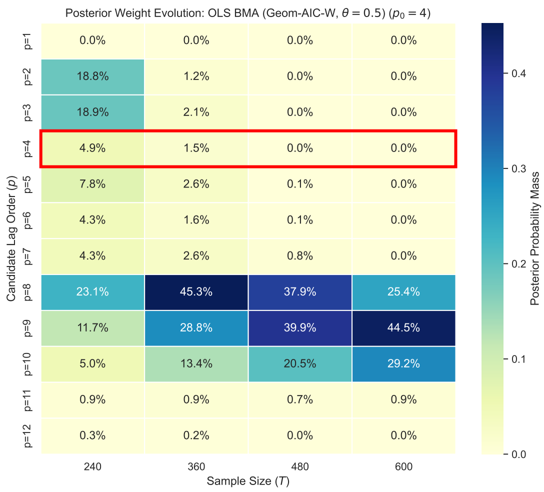
(a) OLS BMA (BIC-weighted)



(b) OLS BMA (Geom. BIC-weighted, $\theta = 0.5$)

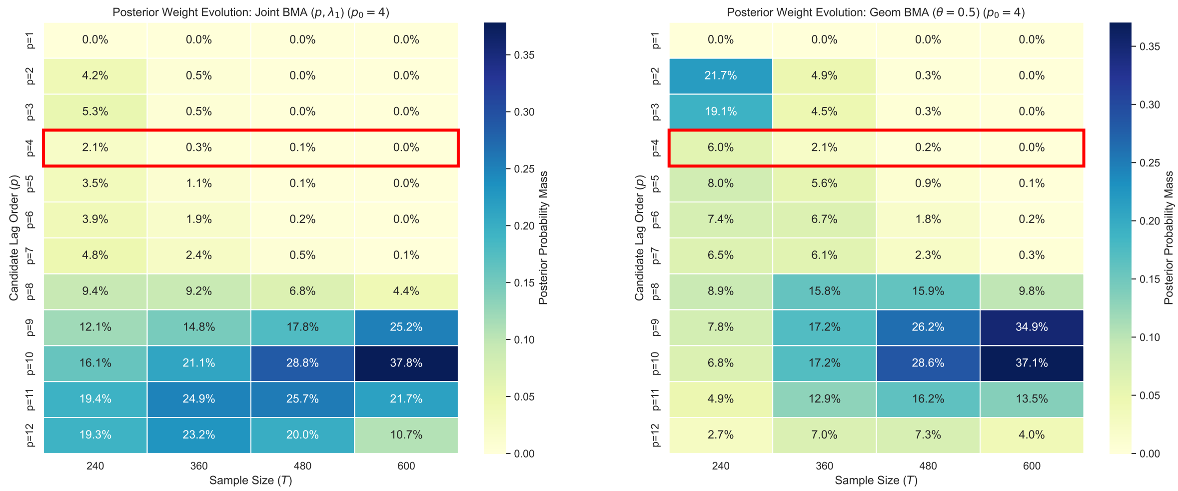


(c) OLS BMA (AIC-weighted)



(d) OLS BMA (Geom. AIC-weighted, $\theta = 0.5$)

Figure 6: Posterior Weight Evolution across the four OLS BMA architectures ($p_0 = 10$).



(a) Joint Grid BMA (b) Geometric Grid BMA

Figure 7: Posterior Weight Evolution for the two BVAR-based BMAs ($p_0 = 10$).

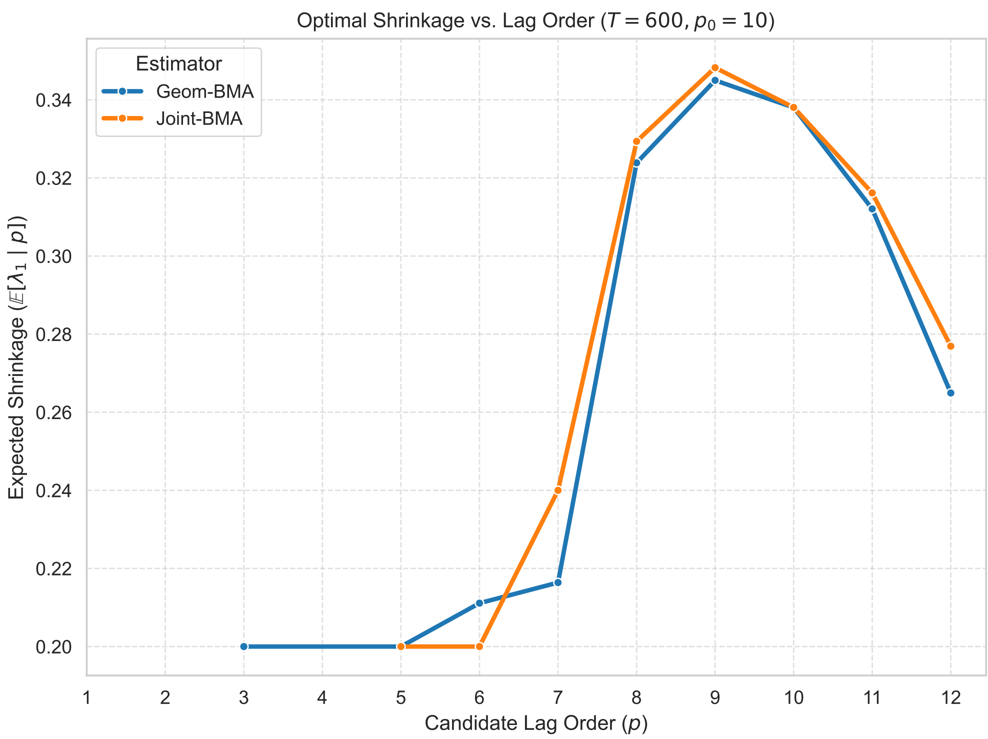


Figure 8: The expected weight of λ_1 given p changes with p ($p_0 = 10$).